



MINHAJ ISLAMIC MODEL SCHOOL HAZRO
FIRST TERM EXAMINATION (2026)

Class: 10th

Subject: Mathematics

Total Marks: 50

Roll No: _____

Student Name: _____

Time Allowed: 2 Hours

SECTION A: OBJECTIVE TYPE (10 Marks)

Q1. Choose the correct option for each question:

(i) $i^2 + i^4 =$

- [A] -1 [B] 0 [C] 1 [D] 2

(ii) Real part of $(2 - 3i)(2 + 3i)$ is:

- [A] -3 [B] 1 [C] 4 [D] 13

(iii) Imaginary part of $(2 - i)(2 + i)$ is:

- [A] 0 [B] 1 [C] 7 [D] 9

(iv) $x + iy$ will be pure imaginary number, when:

- [A] $y = 0$ [B] $x = 0$ [C] $i = 0$ [D] $x = 0, y = 0$

(v) What is additive inverse of $5 - 2i$?

- [A] $5 + 2i$ [B] $-5 - 2i$ [C] $5 - 2i$ [D] $-5 + 2i$

(vi) What is multiplicative inverse of $z = 1 + i$?

- [A] $1 - i$ [B] i [C] $1/2 - 1/2 i$ [D] $-1/\sqrt{2} + 1/\sqrt{2} i$

(vii) If $z = 4 - 3i$, then $z \bar{z} =$

- [A] 3 [B] 9 [C] 16 [D] 25

(viii) Conjugate of $9 - 4i$ is:

- [A] $-9 - 4i$ [B] $9 + 4i$ [C] $9 + 9i$ [D] $4 - 9i$

(ix) If $z = 4 + 4i$, then $z + \bar{z} =$

- [A] 8 [B] $8 + 8i$ [C] $8i$ [D] 0

(x) If $z = 5 + 4i$, then $|z| =$

- [A] 9 [B] 25 [C] 41 [D] $\sqrt{41}$

SECTION B: SHORT ANSWER QUESTIONS ($10 \times 3 = 30$ Marks)

Attempt TEN questions from this section. Each question carries 3 marks.

Q2. Simplify the following:

$$(i^4 + i^3 + i^2 + i)^2$$

Q3. Write in terms of i .

$$\text{i. } 3 - \sqrt{-7} \quad \text{ii. } \sqrt{2} - \sqrt{-3}$$

Q4. Find the values of x and y .

$$(2 + i)x + (1 - 2i)y = 3 + 4i$$

Q5. Write three properties of complex number

Q6. Simplify and write in the form $a + bi$:

$$(3 - 5i) \div (2 - 4i)$$

Q7. If $z_1 = 2 + 5i$, $z_2 = 1 - 3i$ and $z_3 = 2 + i$, then verify that:

$$z_1 z_2 = z_2 z_1$$

Q8. If $z = 5 - 2i$, then verify that

$$|z| = |\bar{z}|$$

Q9. Find the real and imaginary parts of the following complex numbers:

$$\text{(i) } (8 - 3i)^2 \quad \text{(ii) } (5 + 3i)^{-1}$$

Q10. State the condition for two complex numbers to be equal.

Q11. What is the result of multiplying a complex number by its conjugate?

SECTION C: DESCRIPTIVE / LONG QUESTIONS ($2 \times 5 = 10$ Marks)

Attempt TWO (2) questions. Each question carries 5 marks.

Q12. If $(1+i)^2 / (2 - i) = x + iy$, then find the values of x and y .

[5.0 Marks]

Q13. Solve the following simultaneous linear equations with complex coefficients for w and z :

$$z + w = 3i$$

$$2z + 3w = 2$$

[5.0 Marks]

GOOD LUCK☺.